It's Not the Ranker, It's the Data

How retrieval and ranking choices shape a two-stage product recommender (RQ1–RQ3)

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Amazon Reviews 2023, Cell Phones & Accessories · 381k users · 111k products · 2.75M reviews

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The choices around the ranker (which retrievers build the candidate list, how the ranker's training lists and statistics are built) decided final quality far more than the ranker model. Fixing them improved held-out Test nDCG@10 by **61%**.

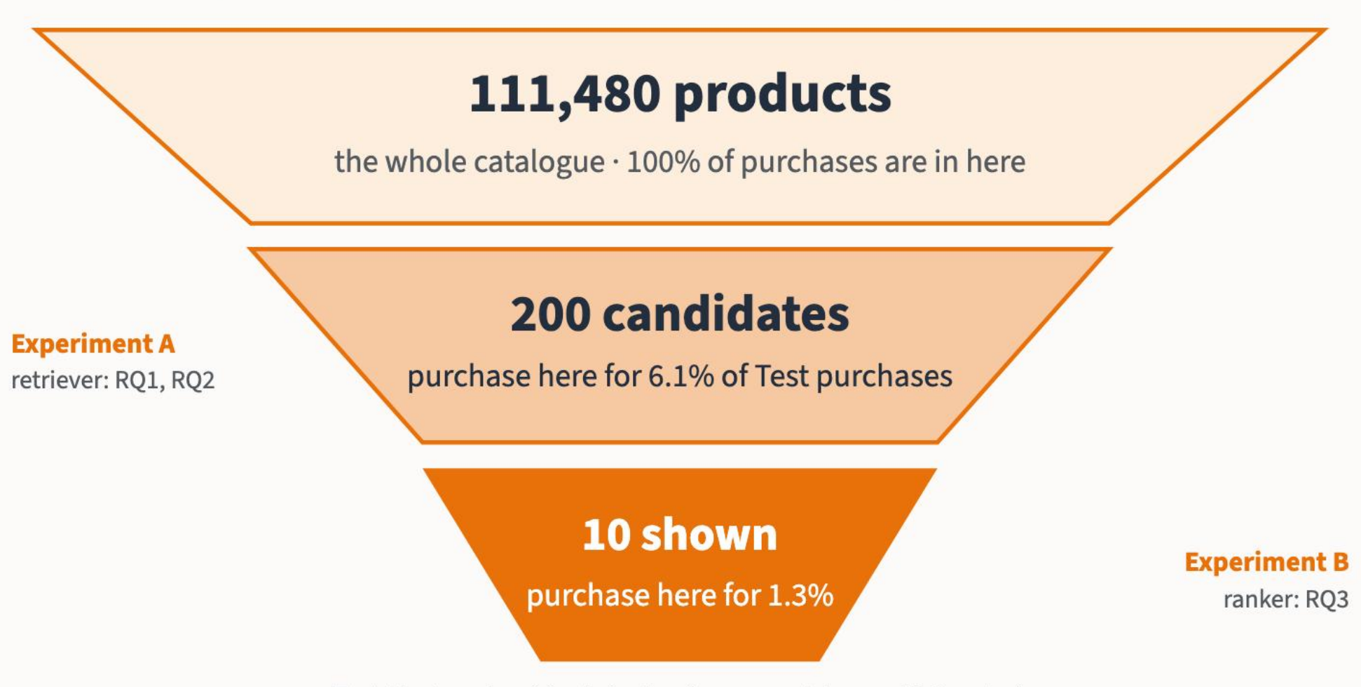
HOW TO READ THIS POSTER · TWO DIMENSIONS

Planned question ↓ · what we found →	**Shortcut 1** **forced training lists**	**Shortcut 2** **ranker learns the retriever**	**Shortcut 3** **statistics age**	**Shortcut 4** **recommends what you own**
RQ1 Retriever design	·	**Two retrievers share 12/200;** mixing +22% Test	·	**Recency** anchors on your own history
RQ2 K and search	**Explains the K curve:** collapse was a training artefact	·	·	·
RQ3 Ranker choice	**Natural lists** +24%; DIN then beats MLP	**Retrain** on the lists it will rank	**Same frozen model** +21% Test	**Ranker cannot add** what retrieval missed
Output Trade-offs and misses	·	·	·	**Who it works for:** repeat buyers, known stores

Rows are the questions we set out to answer (critique slides): panels 1–5. Columns are the mechanisms we found while answering them: the shortcut cards and panel 6.

1 TASK · PIPELINE · DESIGN

A two-stage recommender, studied one stage at a time



Test, the team's original pipeline (recency retriever + MLP ranker)

- **Output:** Recall@K, nDCG@10 and latency trade-offs; retrieval misses vs ranking misses (panel 5).
- **Protocol:** time split (train to Aug 2021, Validation to Jul 2022, then Test); 3 seeds; paired user bootstrap; every choice made on Validation; Test opened once with predictions written first.

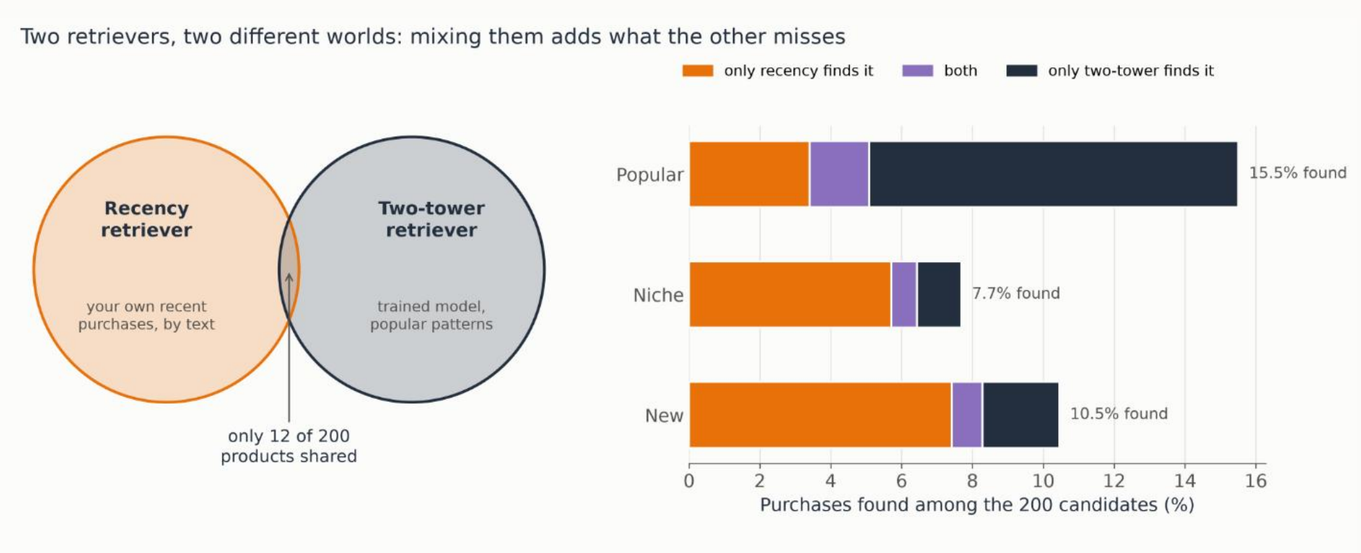
2 RQ1 · EXPERIMENT A · RETRIEVER DESIGN

The trained two-tower beats every baseline, and mixing it with a recency retriever beats both

At a fixed candidate count, does our two-tower retriever beat popularity, BPR-MF, ItemKNN and SASRec on Recall@K?

Retriever, no ranker	Val Recall@200	Val n
Popularity	0.027	
ItemKNN	0.008	
BPR-MF	0.027	
SASRec	0.040	
Two-tower (text)	0.069	
Recency (no training: your recent purchases, by text)	*0.055*	*0.*
Mixed: recency + two-tower, alternating	**0.080**	

Two retrievers, two different worlds: mixing them adds what the other misses



20,000 random Validation purchases. Two-tower finds popular products; recency finds new and niche ones.

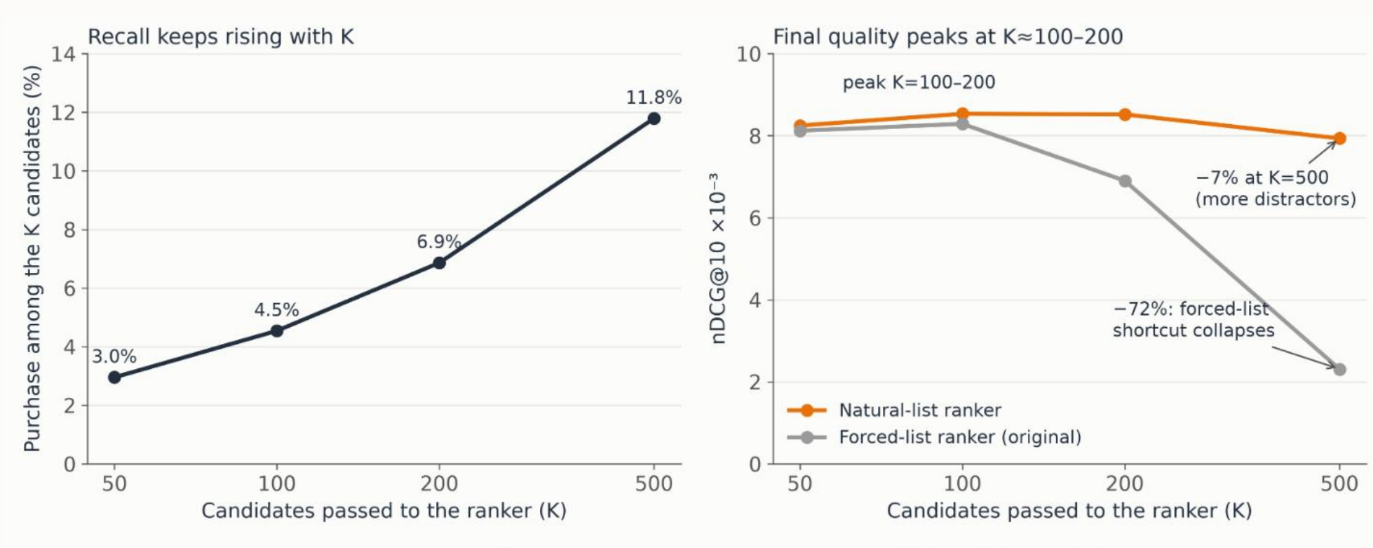
The two find different products: two-tower finds popular ones, recency finds new and niche ones. Only the text-based models find new products at all. With a ranker on top, mixing gives +35% Val, +22% Test.

Baselines: same training period (RQ1 table). Recency and mixed rows: Brandon's pools (P1, P3; alternating order), Validation.

3 RQ2 · EXPERIMENT A · CANDIDATE COUNT AND SEARCH

Recall keeps rising with K, but final quality peaks at K≈100–200

With retriever and ranker fixed, how do K and Flat / HNSW / IVF search affect Recall@K, nDCG@10 and latency?



Validation, two-tower (text) candidates, MLP rankers, mean of 3 seeds. K=500: natural vs its K=100 peak −0.00060 [−0.00079, −0.00041].

Beyond K=200 the extra candidates are mostly distractors: recall nearly doubles to K=500, but the natural-list ranker loses 7%. The forced-list ranker collapses (Shortcut 1).

Index (K=200)	Keeps of exact top 200	Purchases found	nDCG@10 ×10^{-3}	Median latency
Flat (exact)	100%	5.67%	7.37	3.4 ms
HNSW	**99.4%**	**5.67%**	**7.37**	**1.5 ms**
IVF	94.6%	5.47%	6.75	0.9 ms

Why IVF is worse: IVF splits the catalogue into 1,024 clusters and searches only the 32 nearest. Products in other clusters are never looked at, so 5% of the exact top 200 disappear, and some of them are the purchase (5.67% → 5.47% found). HNSW walks a neighbour graph instead and keeps 99.4%, so it is free speed. A lost purchase can't be recovered by the ranker, which is why a 3.5% loss in found purchases costs 8% of nDCG. (1,000-query sample.)

4 RQ3 · EXPERIMENT B · RANKER CHOICE

DIN beats the MLP, but what the ranker is fed matters more than its architecture or loss

On identical lists, how do popularity order, retrieval order, an MLP and DIN compare?

Test nDCG@10 ×10^{-3}	Recency lists	Two-tower lists	Mixed lists
Popularity order	1.51	–	–
Retrieval-score order	6.04	–	–
MLP	6.53	5.19	7.31
DIN	**–**	**6.37**	**8.27**

- **Learned rankers beat simple orderings:** MLP 6.53 vs retrieval order 6.04 vs popularity 1.51 on the same lists.
- **DIN vs MLP** on identical mixed lists: +13% on Test [+0.0008, +0.0011], winning on popular, niche and new products; +23% on two-tower lists.
- **Changing the ranker's loss** (BCE → pairwise) moved Test nDCG@10 by at most 0.0005.
- **Changing what it is fed** moved it far more: natural lists +24%, retraining on mixed lists +35%, statistics of the right age +21% (Shortcuts 1–3 below).

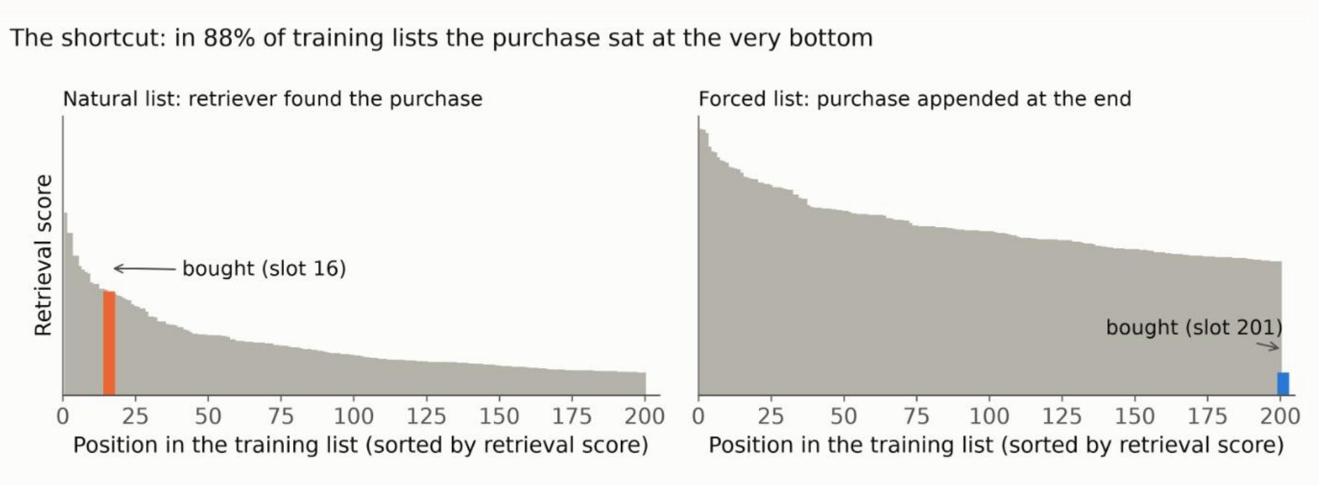
S WHAT WE FOUND WHILE ANSWERING THE RQS · FOUR SHORTCUTS

The ranker learns whatever its inputs make easy

SHORTCUT 1 · EXPLAINS RQ2'S K CURVE, RQ3

Forcing missed purchases into training lists taught "the lowest score is the answer"

The shortcut: in 88% of training lists the purchase sat at the very bottom



Two real ranker-training lists. 276,321 of 315,114 lists were forced; the appended purchase scored below every retrieved candidate in 100% of them.

In 88% of training lists the purchase was appended below every real candidate.

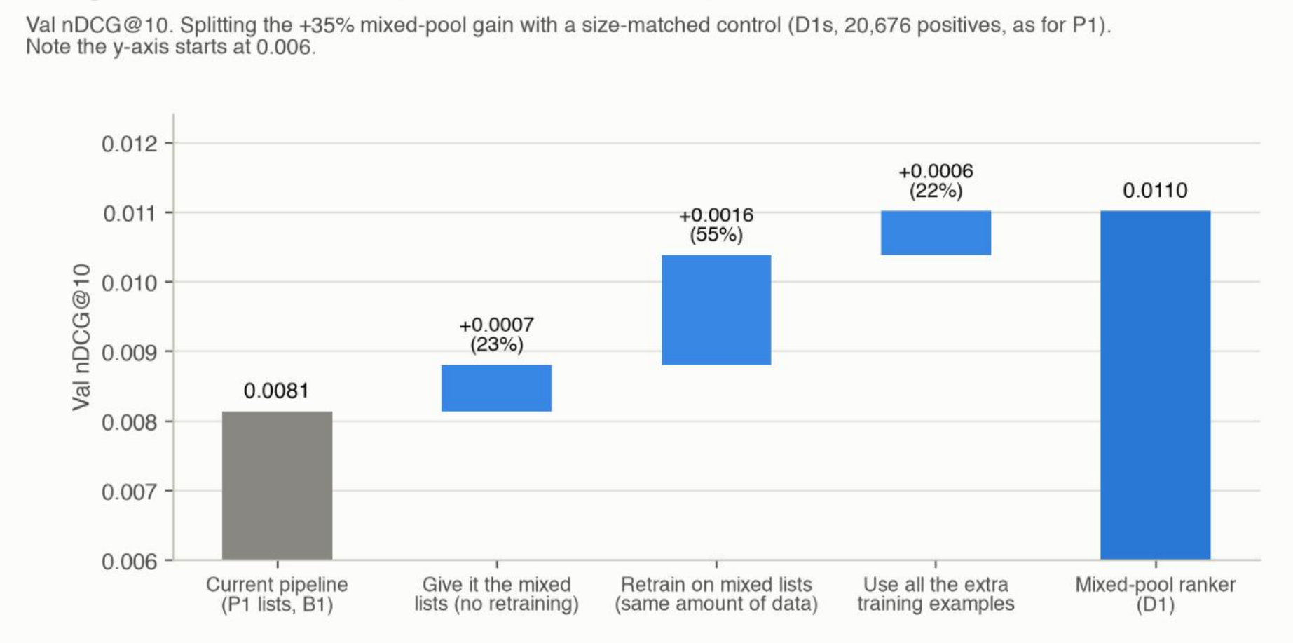
- Natural-list training: +24% at K=200, and no collapse at K=500 (−72% before).
- Two earlier project conclusions ("more candidates hurt", "MLP beats DIN") were artefacts of this.

SHORTCUT 2 · RQ1, RQ3

The ranker learns the habits of the retriever that feeds it

The gain needs the ranker to learn on the lists it will rank

Val nDCG@10. Splitting the +35% mixed-pool gain with a size-matched control (D1s, 20,676 positives, as for P1). Note the y-axis starts at 0.006.



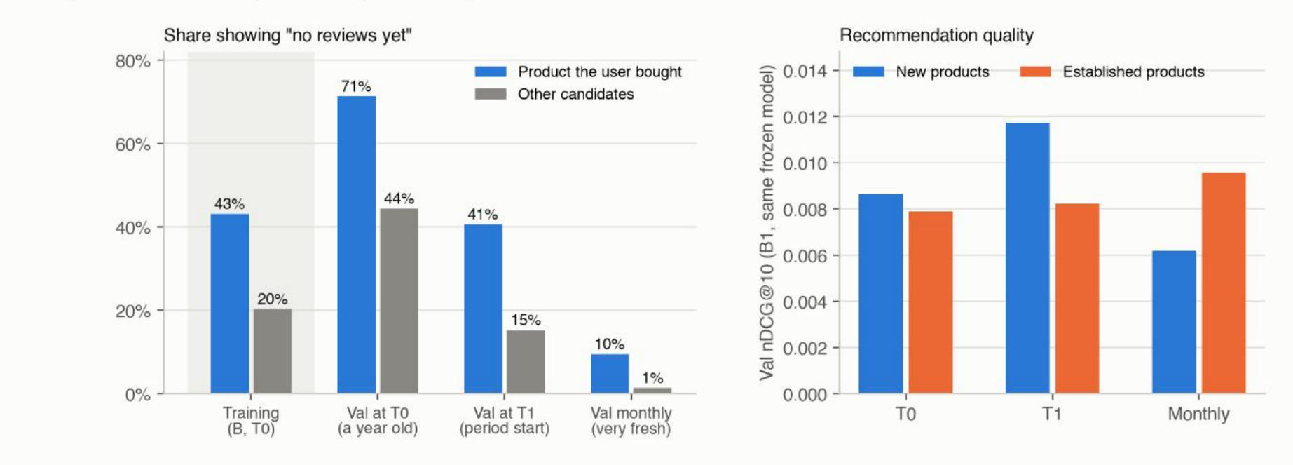
Of the +35% mixed-list gain, only 23% comes from the better candidates alone; 55% needs the ranker retrained on the mixed lists.

SHORTCUT 3 · RQ3

"No reviews yet = likely purchase" only works with statistics as old as in training

The ranker learned "no reviews yet = likely purchase", and it only works with statistics as old as in training

Left: at T1 the cue looks as it did in training; at T0 it fires too often; monthly statistics erase it. Right: the same frozen ranker gains most at T1; monthly statistics help established products but hurt new ones.



In training, 43% of purchases had no reviews vs 20% of other candidates. Statistics recomputed at the period start: +21% on Test, same frozen model. Consistent monthly statistics then make "bought together" inputs useful.

SHORTCUT 4 · RQ1, RQ3, OUTPUT → PANEL 6

The system recommends more of what you already own

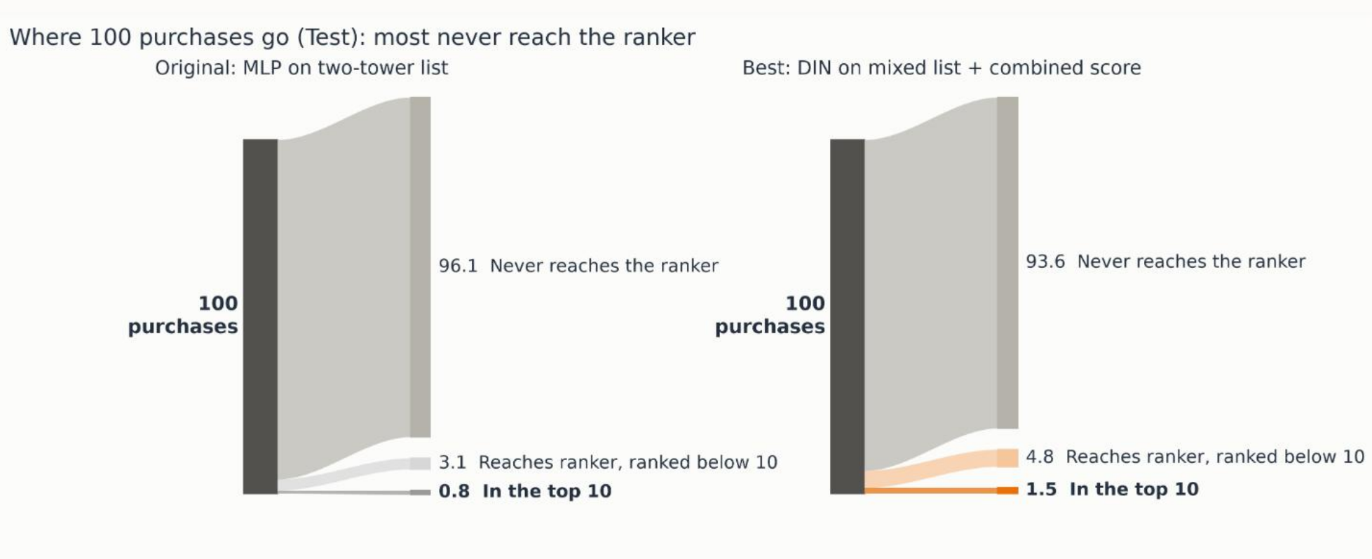
Both retrievers build the query from your own history. Gains concentrate on repeat accessorising from stores you know; upgrades, platform switches and new stores stay unsolved.

- Same-store purchases: 6.6% → 14.7% in the top 10; the other 91%: no gain.
- Upgrades: 37% of candidates are for the old phone, 3% for the new one.
- The ranker shifts slots toward the right phone (26% → 33%) but cannot add what retrieval missed.

5 OUTPUT · EVALUATION

Most failures happen before ranking; the fixed system is 59–61% better for 14–28 ms more per user

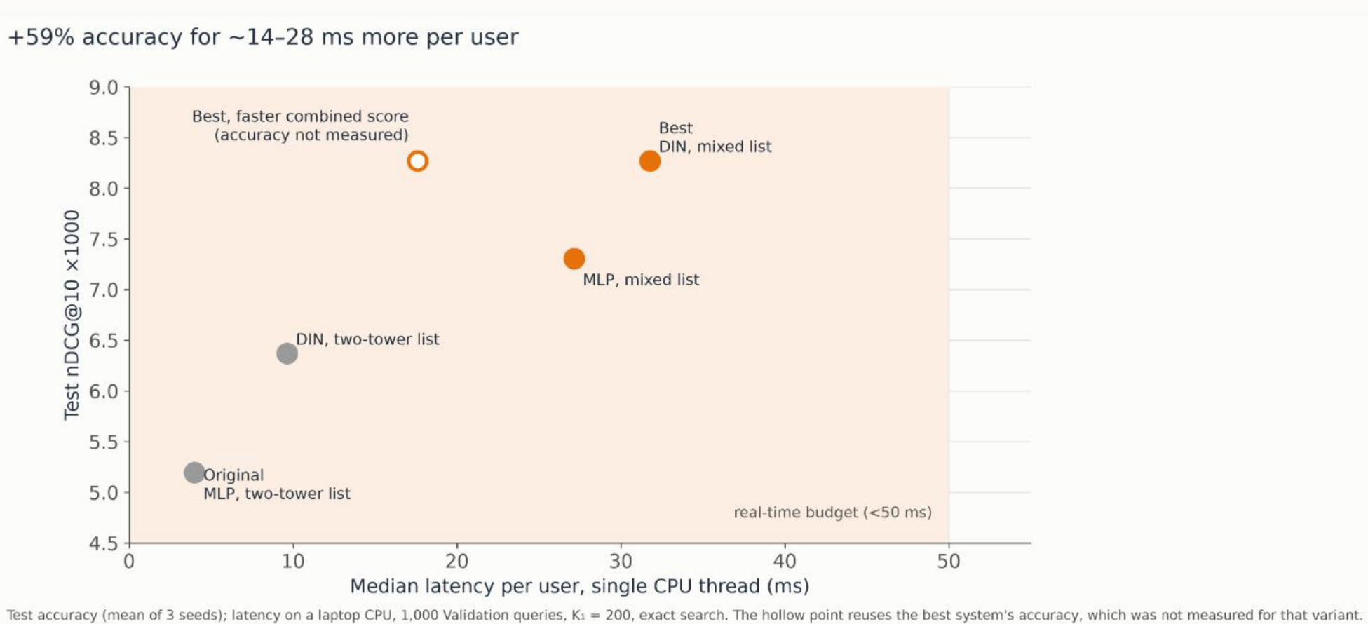
Where 100 purchases go (Test): most never reach the ranker



Test period, 297,390 purchases, mean of 3 seeds. Small groups drawn at a minimum visible height; numbers are exact.

Retrieval misses vs ranking misses on Test.

+59% accuracy for ~14–28 ms more per user



Test accuracy (mean of 3 seeds); latency on a laptop CPU, 1,000 Validation queries, K_1 = 200, exact search. The hollow point reuses the best system's accuracy, which was not measured for that variant.

Accuracy vs latency (laptop CPU, one thread, K=200). Hollow point: faster combined score, accuracy not yet measured.

- **Best combined system** (mixed lists + consistently timed monthly statistics + "bought together" inputs): Test nDCG@10 0.0064 → 0.0104 (+61%); 8 of 9 pre-registered predictions held.
- **DIN on mixed lists** vs the original MLP: 0.0052 → 0.0083 (+59%); new products ×5.8; 4 of 5 predictions held fully.

6 MESO AND MICRO · WHO IT WORKS FOR (SHORTCUT 4 IN DETAIL)

The good, the bad and the ugly: three real buyers

▨ Original (MLP, two-tower list) 🟧 Best (DIN, mixed list) Bars: share of that group's purchases, Validation, 3 seeds

The good · staying in the iPhone 13 family

Owned: iPhone 13 → **Now: iPhone 13 Pro**

Owned a UAG case for the iPhone 13; bought the same UAG case for the 13 Pro (Apr 2022).

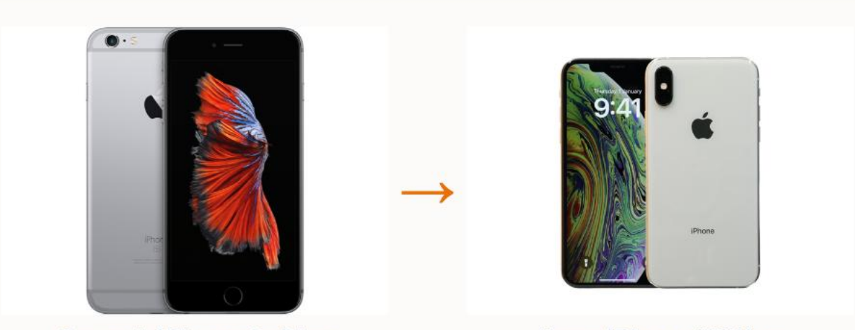
Original	Best
not in its 200	#1

The whole group · 23% of purchases · n=53,486

	Original	Best
Reaches ranker	11.8%	16.5%
In top 10	2.6%	4.5%

Same store, same phone family: the recency retriever finds it, DIN ranks it first.

The bad · upgrading iPhone 6s Plus → XS Max



Owned: iPhone 6s Plus → **Now: iPhone XS Max**

Owned a case and protector for the 6s Plus; bought a leather case for the XS Max (Mar 2022).

All 10 closest candidates	Both systems
old phone	never reached

The whole group · 20% of purchases · n=46,218

	Original	Best
Reaches ranker	5.5%	5.9%
In top 10	1.3%	1.8%

The history pulls the list toward the phone being replaced: 37% of candidates old phone, 3% new.

The ugly · switching Galaxy S20 → iPhone 11

Owned: Galaxy S20 → **Now: iPhone 11**

Owned S20 cases and an ArmorSuit protector; bought an ArmorSuit protector for the iPhone 11 (Aug 2021). Same brand, still missed.

Original	Best
never reached	never reached

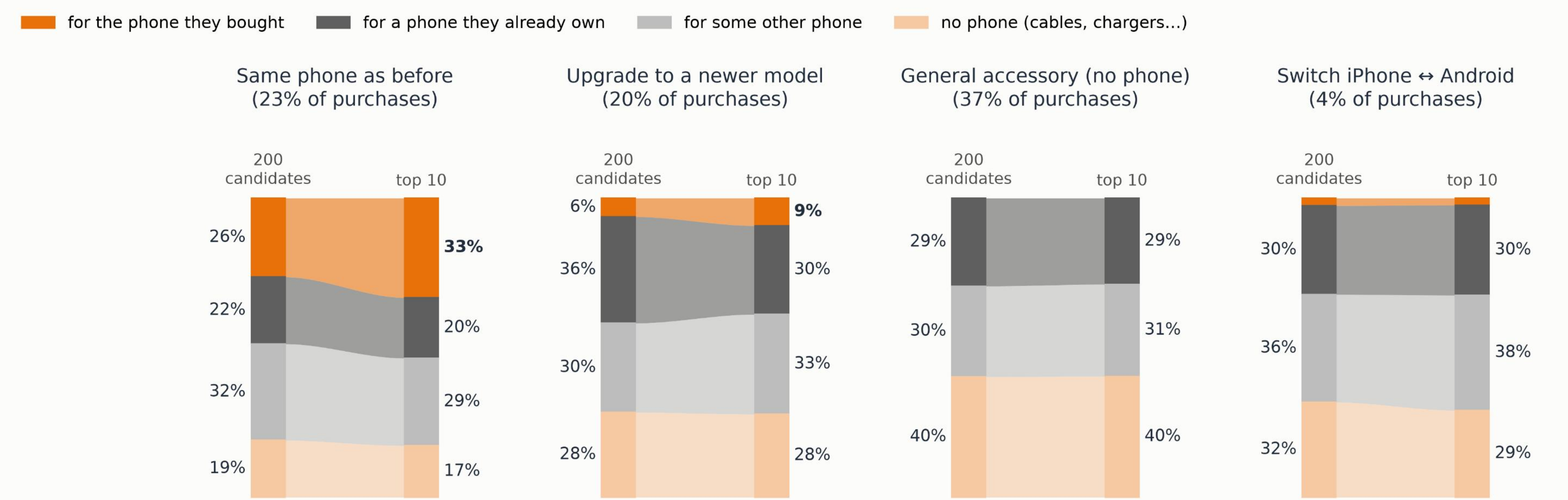
The whole group · 4% of purchases · n=9,117

	Original	Best
Reaches ranker	1.5%	1.6%
In top 10	0.2%	0.3%

Not an upgrade (the iPhone 11 is older than the S20): a second phone, a family member's, or a switch. Nothing in a Samsung history points to an iPhone, so neither retriever looks there.

Three real Validation purchases with their Amazon product photos; phone photos are illustrations of the models named in the product titles. The bars under each are the measured result for all purchases of that kind (phone labels from titles, approximate). Product images: Amazon; phone images: Apple and retailer sites.

What fills the list and the top 10: the ranker moves slots toward the right phone, but cannot add what retrieval missed

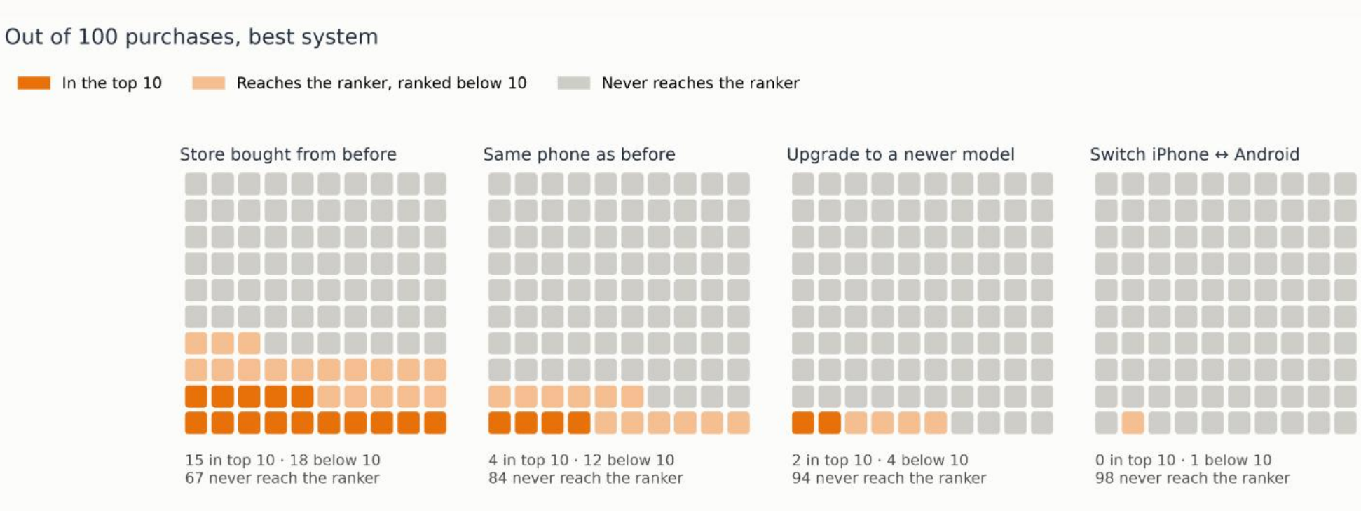


Same phone as before	Upgrade to a newer model	General accessory (no phone)	Switch iPhone ↔ Android
Top 10 has ≥1 for the bought phone: 81%	Top 10 has ≥1 for the bought phone: 48%	Same product type: 23% → 22%	Top 10 has ≥1 for the bought phone: 16%
Same product type: 35% → 35%	Same product type: 28% → 27%	Purchase reaches the ranker: 7.3%	Same product type: 26% → 26%
Purchase reaches the ranker: 17.1%	Purchase reaches the ranker: 6.4%	Exact purchase in top 10: 1.8%	Purchase reaches the ranker: 1.2%
Exact purchase in top 10: 4.6%	Exact purchase in top 10: 2.0%		Exact purchase in top 10: 0.4%

Best system (DIN on mixed list + combined score), 20,000 random Validation purchases, seed 0. Slots classified by the phone models product titles name (approximate).

Right phone, wrong product: for same-phone buyers 81% of top 10s hold something for their phone, but only 4.6% hold the exact purchase. The ranker shifts slots toward the right phone but cannot add what retrieval missed.

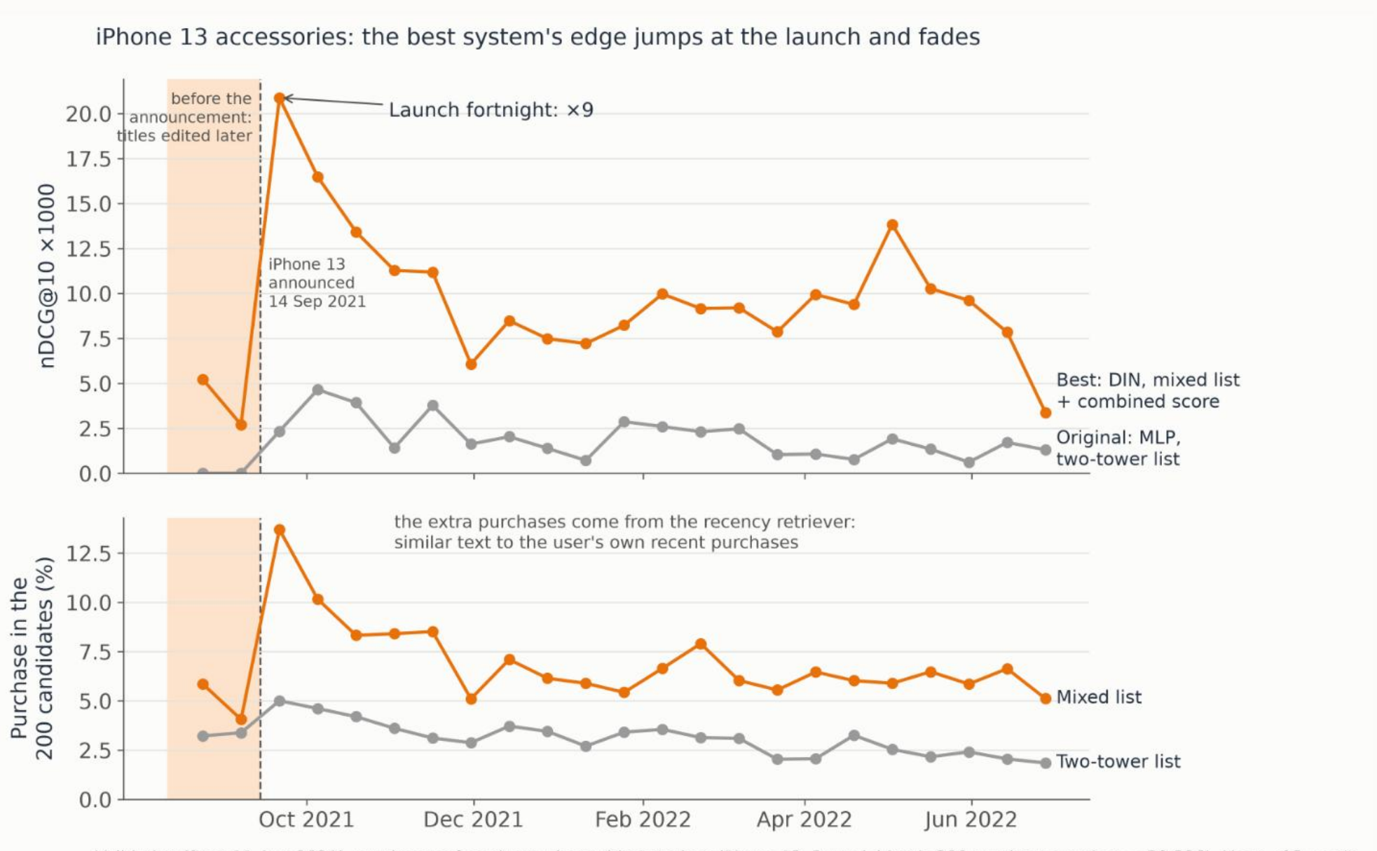
Out of 100 purchases, best system



Validation, mean of 3 seeds; each square = 1 purchase in 100 (rounded). Labels from titles (approximate).

Store loyalty carries the gain: 15 in 100 same-store purchases reach the top 10; platform switchers 0.

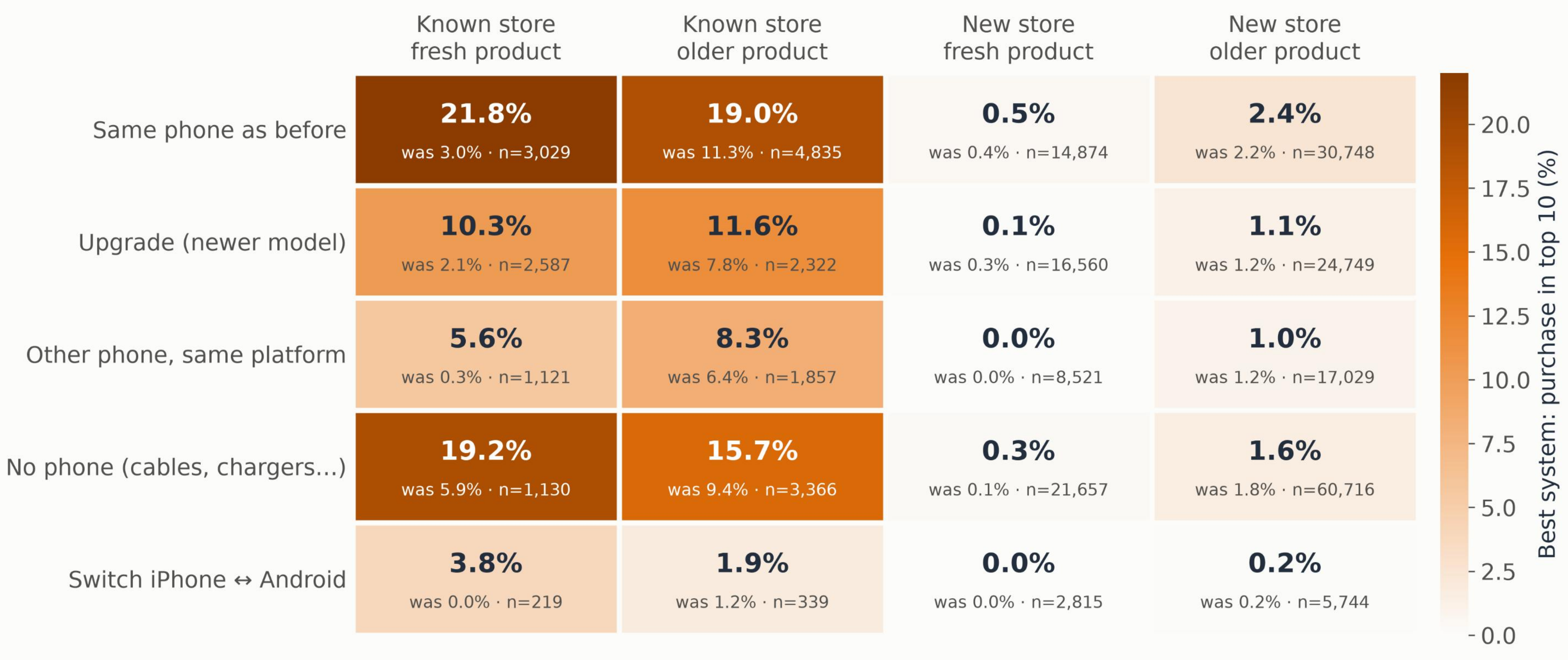
iPhone 13 accessories: the best system's edge jumps at the launch and fades



Validation (from 11 Aug 2021), purchases of products whose title mentions iPhone 13, 2-week bins (≥200 purchases each; n = 29,386). Mean of 3 seeds.

Launch wave: iPhone 13 launch fortnight ×9, fading to about ×4 by December. Same for Pixel 6, iPhone SE 2022, MagSafe; Apple Watch bands are the exception.

User × product: the sweet spot is a fresh product from a known store, for a phone you own



	Known store fresh product	Known store older product	New store fresh product	New store older product
Same phone as before	**21.8%** was 3.0% · n=3,029	**19.0%** was 11.3% · n=4,835	**0.5%** was 0.4% · n=14,874	**2.4%** was 2.2% · n=30,748
Upgrade (newer model)	**10.3%** was 2.1% · n=2,587	**11.6%** was 7.8% · n=2,322	**0.1%** was 0.3% · n=16,560	**1.1%** was 1.2% · n=24,749
Other phone, same platform	**5.6%** was 0.3% · n=1,121	**8.3%** was 6.4% · n=1,857	**0.0%** was 0.0% · n=8,521	**1.0%** was 1.2% · n=17,029
No phone (cables, chargers...)	**19.2%** was 5.9% · n=1,130	**15.7%** was 9.4% · n=3,366	**0.3%** was 0.1% · n=21,657	**1.6%** was 1.8% · n=60,716
Switch iPhone ↔ Android	**3.8%** was 0.0% · n=219	**1.9%** was 1.2% · n=339	**0.0%** was 0.0% · n=2,815	**0.2%** was 0.2% · n=5,744

Validation, best system (DIN, mixed list + combined score); 'was' = original system. Mean of 3 seeds. Phone labels from titles (approximate); small cells are noisy.

Sweet spot: fresh product, known store, phone you own → 21.8% in the top 10 (was 3.0%).

Takeaways

1. **RQ1:** the trained two-tower beats every baseline; two retrievers that see different products beat any one.
2. **RQ2:** recall rises with K but quality peaks at K≈100–200; HNSW is free speed, IVF drops purchases the ranker can't recover.
3. **RQ3:** DIN beats the MLP, yet natural training lists, retraining on the right lists and statistics of the right age matter more.
4. **Output:** most misses are retrieval misses; the system serves buyers accessorising a phone they own.

Every shortcut is one lesson: the ranker learns whatever its inputs make easy.

LIMITATIONS

The data remembers the future: sellers edit titles to add new phones

770

purchases made before the iPhone 13 was announced (14 Sep 2021) have “iPhone 13” in the product title

bought 2021-09-08
KISEN [5 in 1] for **iPhone 13** Pro Max Case, with 2 Pack Tempered Glass Screen Protector + 2 P...

bought 2021-08-23
AkoaDa 3 Pack **iPhone 13** Charger Adapter, Type C to USB Adapter, iPhone 12 Charger Adapter Co...

bought 2021-08-15
iPhone 13 Pro Case Square,Grey Marble Square Design iPhone 13 Pro Cases for Girl Women,Slim ...

bought 2021-08-13
ANDATE Black **iPhone 13** Pro Max Silicone Case, Compatible with iPhone 13 Pro Max Full Body Ca...

Validation. Product titles are today's metadata, not as of purchase, so title-based labels (and text features) carry some future information.

Product titles are today's metadata: sellers add new phones later. Phone labels (and text features) carry some future information.

Validation unless marked Test. Phone and platform labels from titles are approximate. Latency measured on a laptop CPU, single thread.